

Solturio - File Storage & Hash Architecture

Four-Tier Storage System

Layer	What is Stored	Purpose
PostgreSQL	Thumbnails, metadata, ownership claims	Primary DB, fast queries
IPFS (Pinata)	Metadata JSON with file hashes	Decentralized storage
Arweave	Verified badge images (gold check)	Permanent storage
Solana	NFT certificates with hashes	Proof of ownership

Hash Flow

1. User uploads file - Sharp extracts metadata (dimensions, format, color)
2. SHA-256 hash generated from file content
3. Metadata JSON created with hash, ownership claim, timestamp
4. No full file stored - only hash and metadata (user keeps original)
5. Hash stored in PostgreSQL for quick verification lookups
6. Metadata pinned to IPFS via Pinata
7. NFT minted on Solana with hash in on-chain metadata

Verification Logic

- Extension/API receives file and generates SHA-256 hash
- Hash compared against PostgreSQL fileHash column
- If match found: content is registered, return owner info
- If no match: content not registered

Key Points

- We do not store actual files - just metadata and hashes
- User is responsible for keeping originals
- Hash = proof of content (same content = same hash)
- Gold Check Badge - visual overlay stored on Arweave

This approach minimizes storage costs while maintaining cryptographic proof of ownership.